Enrollment No._____

GANPAT UNIVERSITY

B.Sc.IT(DS) Sem-III Reg\Rem Nov-Dec 2024

U73A1DS DATA STRUCTURE

Time: 3 Hours

Total: 60 Marks

Instructions:

1. Figures to the right indicate full marks, Course Outcomes and BTL Pattern Level.
2. This Question paper has two sections. Each section should be written in a separate answer book.
3. Section-I consist of 07 questions of 6 marks each with options respectively. Section-II consist of 03 questions, Question 2 of 12 Marks, Question 3 of 12 Marks and Question 4 of 6 Marks with options respectively.
4. Be precise and to the point in your answer.
5. CO is Course Outcome from the Subject Syllabus.
6. Bloom's Taxonomy Based Assessment Pattern: BTL1-Remember; BTL2-Understand; BTL3- Apply; BTL4-Analyze; BTL5- Evaluate; BTL6-Create.

Section – I

Q-1	Answer the following. (Any 5)	Marks 30	CO	Bloom's Taxonomy
1	Write down algorithm to perform insert operation at specific position on singly Linked List.	06	CO3	BTL3
2	Write down algorithm to perform insert and delete operation on double ended queue.	06	CO2, CO3	BTL3
3	Explain graph representation techniques.	06	CO3	BLT2
4	Sort the following array using Bubble sort: [10, 7, 8, 9, 1, 5] Show each step of the process.	06	CO4	BTL1, BLT2
5	Write down algorithm to perform delete operation at first node on doubly Linked List.	06	CO2, CO3	BTL3
6	What is a Data Structure? Explain different types of data structures, their uses, and provide examples.	06	CO1	BTL1, BTL4
7	What is Stack? Write down algorithm for stack operations.	06	CO2	BTL2, BTL3

Section – II

Q-2	Answer the following (Any Two)	Marks 12	CO	Bloom's Taxonomy
1	Explain Polish Notation (Prefix Notation) and Reverse Polish Notation (Postfix Notation). How are they evaluated? Illustrate with examples.	06	CO1 CO2	BTL5
2	Write down algorithm to perform insert operation at last node on doubly Linked List.	06	CO2, CO3	BTL3
3	Write down algorithm for insert and delete operation on circular queue.	06	CO2, CO3	BTL3

Q.3	**Answer the following: (Any Two)**	**12**		
1	What is a binary search tree (BST)? Define its properties. Given the following sequence of numbers, construct a binary search tree: [24, 15, 50, 10, 22, 35, 75]	06	CO3	BTL1, BTL3
2	What are the different types of tree traversal techniques? List them and explain any one with suitable example.	06	CO3	BTL1, BTL4
3	Define the term: Graph Traversal. Also, write down algorithm for any one graph traversal technique.	06	CO2	BTL2, BTL3
Q.4	**Answer the following: (Any One)**	**06**		
1	Sort the following array using quicksort: [10, 7, 8, 9, 1, 5] Show each step of the process.	06	CO4	BTL3
2	Sort the array [28, 10, 14, 37, 12] using selection sort and show each step of the sorting process.	06	CO4	BTL3

**** End of the Paper ****

Enrollment No._____

GANPAT UNIVERSITY

B. Sc.IT (DS) Sem-III Reg/Rem Nov-Dec 2024

U73A2ADS ADVANCE DATABASE MANAGEMENT SYSTEM

Time: 3 Hours

Total: 60 Marks

Instructions:

1. Figures to the right indicate full marks, Course Outcomes and BTL Pattern Level.
2. This Question paper has two sections. Each section should be written in a separate answer book.
3. Section-I consist of 07 questions of 6 marks each with options respectively. Section-II consist of 03 questions, Question 2 of 12 Marks, Question 3 of 12 Marks and Question 4 of 6 Marks with options respectively.
4. Be precise and to the point in your answer.
5. CO is Course Outcome from the Subject Syllabus.
6. Bloom's Taxonomy Based Assessment Pattern: BTL1-Remember; BTL2-Understand; BTL3- Apply; BTL4-Analyze; BTL5- Evaluate; BTL6-Create.

Section – I				
Q-1	**Answer the following. (Any 5)**	**Marks 30**	**CO**	**Bloom's Taxonomy**
1	List the advantages of using PL/SQL in database management.	06	CO1	BTL1
2	Explain the structure of a PL/SQL block and its components.	06	CO2	BTL2
3	Describe the different types of database triggers in Oracle.	06	CO3	BTL2
4	Create a user-defined exception handler for a scenario where a specific error may occur.	06	CO2	BTL3
5	Analyze the advanced feature of data sampling in Data Pump.	06	CO3	BTL4
6	Evaluate the use of SQL*Loader data types in data loading.	06	CO4	BTL5
7	Analyze the components of a PL/SQL package.	06	CO2	BTL4
Section – II				
Q-2	**Answer the following (Any Two)**	**Marks 12**	**CO**	**Bloom's Taxonomy**
1	Define the concept of the two-phase commit protocol.	06	CO2	BTL1
2	Define discretionary access control in the context of database security.	06	CO3	BTL1
3	Write a simple PL/SQL block that calculates the factorial of a number.	06	CO2	BTL3
Q.3	**Answer the following: (Any Two)**	**12**		
1	Define PL/SQL procedures and functions and explain their advantages.	06	CO2	BTL1
2	Provide an overview of Oracle Data Pump.	06	CO4	BTL1
3	Analyze the benefits of using a Directory Object in data movement.	06	CO3	BTL4

Q.4	**Answer the following: (Any One)**	**06**		
1	Define SQL*Loader and its primary purpose.	06	CO4	BTL1
2	Explain the components of an SQL*Loader control file.	06	CO4	BTL2

**** End of the Paper ****

Enrollment No._____

GANPAT UNIVERSITY

B.Sc. IT (Data Science) Sem-III Regular/Remedial Nov-Dec 2024

U73A3OCP OBJECT ORIENTED CONCEPT AND PROGRAMMING

Total: 60 Marks

Time: 3 Hours

Instructions:

1. Figures to the right indicate full marks, Course Outcomes and BTL Pattern Level.
2. This Question paper has two sections. Each section should be written in a separate answer book.
3. Section-I consist of 07 questions of 6 marks each with options respectively. Section-II consist of 03 questions, Question 2 of 12 Marks, Question 3 of 12 Marks and Question 4 of 6 Marks with options respectively.
4. Be precise and to the point in your answer.
5. CO is Course Outcome from the Subject Syllabus.
6. Bloom's Taxonomy Based Assessment Pattern: BTL1-Remember; BTL2- Understand; BTL3- Apply; BTL4-Analyze; BTL5- Evaluate; BTL6-Create.

Section – I				
Q-1	**Answer the following. (Any 5)**	**Marks 30**	**CO**	**Bloom's Taxonomy**
1	Elaborate the type-conversation in details.	06	CO1	BTL5
2	What is constructor? Elaborate the Constructor in detail.	06	CO1	BTL2/BTL5
3	What is Inheritance? Explain types of inheritance in java.	06	CO2	BTL2/BTL4
4	How to apply implementation package?	06	CO3	BTL3
5	State the different between vector and list in Java.	06	CO3	BTL4
6	What is Lambda Expression? How to apply parameters in lambda expression function?	06	CO4	BTL2/BTL3
7	Analyze the Fork/Join using parallel programming.	06	CO5	BTL4
Section – II				
Q-2	**Answer the following (Any Two)**	**Marks 12**	**CO**	**Bloom's Taxonomy**
1	What is polymorphism? Demonstrate the types of polymorphism with example.	06	CO1	BTL2/BTL4
2	What is abstraction? Analyze Abstract class with example.	06	CO2	BTL2/BTL4
3	Demonstrate about try, catch, finally, throw and throws.	06	CO2	BTL3
Q.3	**Answer the following: (Any Two)**	**12**		
1	Elaborate Random class with example.	06	CO3	BTL5
2	Apply to implement exception handling in lambda expression.	06	CO4	BTL3
3	Illustrate method and constructor references in details.	06	CO4	BTL4
Q.4	**Answer the following: (Any One)**	**06**		
1	Illustrate Callable and Future in detail.	06	CO5	BTL4
2	Elaborate the synchronization in details.	06	CO5	BTL5

**** End of the Paper ****

Enrollment No._____

GANPAT UNIVERSITY

B.Sc. IT (Data Science) Sem-III Reg\Rem Nov-Dec 2024

U73A4AS Advance Statistics

Time: 3 Hours | Total: 60 Marks

Instructions:

1. Figures to the right indicate full marks, Course Outcomes and BTL Pattern Level.
2. This Question paper has two sections. Each section should be written in a separate answer book.
3. Section-I consist of 07 questions of 6 marks each with options respectively. Section-II consist of 03 questions, Question 2 of 12 Marks, Question 3 of 12 Marks and Question 4 of 6 Marks with options respectively.
4. Be precise and to the point in your answer.
5. CO is Course Outcome from the Subject Syllabus.
6. Bloom's Taxonomy Based Assessment Pattern: BTL1-Remember; BTL2- Understand; BTL3-Apply; BTL4-Analyze; BTL5- Evaluate; BTL6-Create.

Section – I

Q-1	Answer the following. (Any 5)	Marks 30	CO	Bloom's Taxonomy
1	Round off number 3.1428571 to three significant numbers then compute absolute error, relative error, percentage error.	06	CO1	BTL5
2	Find root of the equation $f(x) = x^3 + x^2 - 1 = 0$ using iteration method upto four decimal places.	06	CO1	BTL3
3	Solve the following system of equation using Jacobi iteration method. (1)$20x + y - 2z = 17$,(2) $3x + 20y - z = -18$,(3) $2x - 3y + 20z = 25$	06	CO2	BTL3
4	An experiment consist of three independent tosses of a fair coin. Let x =the number of heads, y =the number of head runs, z =the length of head runs, a head run being defined as consecutive occurrence of at least two heads, its length then being the number of heads, its length then being the number of heads occurring together in three tosses of coin. Find the probability function of (1) x ,(2)y, (3)z ,(4) $x + y$,(5) xy and construct probability tables and probability charts.	06	CO3	BTL6
5	Two balls are selected at random from a box containing three red, two green and four white. If X and Y are the number of red balls and green balls respectively included among the two balls drawn from the box. (1) Find joint probability of X and Y. (2) Find marginal probability distribution of X and Y .	06	CO3	BTL5
6	What is probability sampling? Explain any two probability sampling methods.	06	CO4	BTL2

7 Fit trend line to the following data by the least squares method. 06 CO5 BT

Years	1985	1987	1989	1991	1993
Production (in tons)	18	21	23	27	16

Section – II

		Marks	CO	Bloom's Taxonomy
Q-2	**Answer the following (Any Two)**	**12**		
1	Find root of the equation $f(x) = 3x - cosx - 1$ using Newton-Raphson method upto four decimal places.	06	CO1	BTL3
2	Solve the following system of equations by Gauss-Jordan elimination method, (1) $x + y - z + w = 0$, (2) $x - y + 2z - w = 0$, (3) $3x + y + w = 0$	06	CO2	BTL3
3	Investigate for what values of λ and μ equation $x + 2y + z = 8$, $2x + 2y + 2z = 13, 3x + 4y + \lambda z = \mu$ have(1) no solution ,(2) unique solution, (3)infinite solution.	06	CO2	BTL5
Q.3	**Answer the following: (Any Two)**	**12**		
1	A random variable x has the following probability function values of x,	06	CO3	BTL4

x	0	1	2	3	4	5	6	7
$P(x)$	0	k	$2k$	$2k$	$3k$	k^2	$2k^2$	$7k^2 + k$

(1) If $P(x \leq a) > \frac{1}{2}$ then find minimum value of a.
(2)Find $P(0 < x < 5)$.

2	Let x I be a continuous random variable with probability density function given by	06	CO3	BTL5

$$f(x) = \begin{cases} ax\,; 0 \leq x \leq 1 \\ a; 1 \leq x \leq 2 \\ -ax + 3a\,; 2 \leq x \leq 3 \\ 0\,; 3 < x \end{cases}$$

(1) Determine constant a.
(2) Calculate $P(x \leq 1.5)$.

3	Explain convenience sampling and Quota sampling.	06	CO4	BTL2
Q.4	**Answer the following: (Any One)**	**06**		
1	Apply least squares techniques to obtain trend values from the data given below.	06	CO5	BTL5

Years	1998	1999	2000	2001
Sales (in tons)	100	120	110	140

2	Calculate the five moving average of the following.	06	CO5	BTL5

Year	1950	1951	1952	1953	1954	1955	1956
Values	105	115	100	90	80	95	85

**** End of the Paper ****

GANPAT UNIVERSITY

B.Sc.IT(Data Science) Sem-III Regular Nov-Dec 2024

U73A5AI INTRODUCTION TO ARTIFICIAL INTELLIGENCE

Time: 3 Hours Total: 60 Marks

Instructions:

1. Figures to the right indicate full marks, Course Outcomes and BTL Pattern Level.
2. This Question paper has two sections. Each section should be written in a separate answer book.
3. Section-I consist of 07 questions of 6 marks each with options respectively. Section-II consist of 03 questions, Question 2 of 12 Marks, Question 3 of 12 Marks and Question 4 of 6 Marks with options respectively.
4. Be precise and to the point in your answer.
5. CO is Course Outcome from the Subject Syllabus.
6. Bloom's Taxonomy Based Assessment Pattern: BTL1-Remember; BTL2-Understand; BTL3- Apply; BTL4-Analyze; BTL5- Evaluate; BTL6-Create.

	Section – I			
Q-1	**Answer the following. (Any 5)**	**Marks 30**	**CO**	**Bloom's Taxonomy**
1	Define Artificial Intelligence (AI). Discuss its key advantages and disadvantages.	06	CO1	BTL3
2	Describe the Hill Climbing algorithm, including its main features.	06	CO2	BTL4
3	What is Knowledge Representation in AI? Discuss the primary techniques used for effective knowledge representation.	06	CO2	BTL5
4	Explain Minimalist Reasoning in AI and its applications.	06	CO3	BTL4
5	Provide a detailed explanation of Semantic Nets, including their structure, components, and applications.	06	CO3	BTL3
6	Discuss Conceptual Dependency in AI, including its goals, structure, and practical applications.	06	CO4	BTL5
7	How can a problem be defined as a State Space Search? Provide an example to illustrate this concept.	06	CO2	BTL5
	Section – II			
Q-2	**Answer the following (Any Two)**	**Marks 12**	**CO**	**Bloom's Taxonomy**
1	Describe the different types of Artificial Intelligence (AI).	06	CO1	BTL3
2	Explain the role of Constraint Satisfaction in problem-solving and how it applies to problems with specific constraints.	06	CO2	BTL4
3	Define Generate-And-Test in detail.	06	CO2	BTL4
Q.3	**Answer the following: (Any Two)**	**12**		
1	Discuss the Is-A relationship for representing instances in knowledge representation.	06	CO3	BTL5
2	Define a frame in AI.	06	CO3	BTL3
3	Explain Logical Connectives in knowledge representation, providing examples to illustrate.	06	CO3	BTL4
Q.4	**Answer the following: (Any One)**	**06**		
1	Discuss Scripts in knowledge representation, including their structure and applications.	06	CO4	BTL4
2	Provide a detailed explanation of the CYC.	06	CO4	BTL3

— END OF PAPER —

Enrollment No._____

GANPAT UNIVERSITY

B.Sc.IT(DATA SCIENCE)- III Reg\ Rem – Nov-Dec. 2024

U73B6EDM: ENVIRONMENT DISASTER MANAGEMENT

Time: 3 Hours **Total: 60 Marks**

Instructions:

1. Figures to the right indicate full marks, Course Outcomes and BL Pattern Level.
2. This Question paper has two sections. Each section should be written in a separate answer book.
3. Section-I consist of 07 questions of 6 marks each with options respectively. Section-II consist of 03 questions, Question 2 of 12 Marks, Question 3 of 12 Marks and Question 4 of 6 Marks with options respectively.
4. Be precise and to the point in your answer.
5. CO is Course Outcome from the Subject Syllabus.
6. Bloom's Taxonomy Based Assessment Pattern: BTL1-Remember; BTL2-Understand; BTL3- Apply; BTL4-Analyze; BTL5- Evaluate; BTL6-Create.

Section – I				
Q-1	**Answer the following. (Any 5)**	**Marks 30**	**CO**	**Bloom's Taxonomy Pattern**
1	How nature balances in between their elements? Explain.	06	CO1	BTL4
2	What is ecosystem? Explain ecosystem within different spheres of earth.	06	CO2	BTL5
3	Explain the Green House Effect and term Global Warming.	06	CO3	BTL4
4	Discuss disaster management cycle.	06	CO4	BTL5
5	Explain the scope of environment.	06	CO1	BTL3
6	Evaluate the reason for acid rain. How it effects on life?	06	CO3	BTL5
7	Illustrate the planning for Earthquakes and Landslides disasters.	06	CO5	BTL3
Section – II				
Q-2	**Answer the following (Any Two)**	**Marks 12**	**CO**	**Bloom's Taxonomy Pattern**
1	What is environment define the term and explain in detail.	06	CO1	BTL4
2	Define the term ecology and explain it.	06	CO2	BTL5
3	Define: Producer, Consumer and Decomposer.	06	CO2	BTL3
Q.3	**Answer the following: (Any Two)**	**12**		
1	Explain the common sources of air pollution? How can we control air pollution?	06	CO4	BTL5
2	Explain the effects on human life due to different pollutions.	06	CO3	BTL3
3	Give the classification of Hazards. List and brief five hazards in each of the category.	06	CO4	BTL5
Q.4	**Answer the following: (Any One)**	**06**		
1	Explain disaster due to nuclear power accidents.	06	CO5	BTL3
2	Write short notes on war and Terrorisms.	06	CO5	BTL3

**** End of the Paper ****